

Ex. A car hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as poisson Variable with mean 1.5. Calculate the proportion of days on which (i) neither car is used and (ii) some demand is refused.

Sol. The proportion of days on which there are  $x$  demands for a car  $= P \{ \text{of } x \text{ demands in a day} \}$   
$$= \frac{e^{-1.5} (1.5)^x}{x!}$$

$$(i) P(X=0) = e^{-1.5} = 0.2231$$

$$\begin{aligned} (ii) P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - [P(X=0) + P(X=1) + P(X=2)] \\ &= 1 - 0.2231 \times 3.675 = 0.19126 \end{aligned}$$



Ex A pin manufacturer knows that 5% of his product is defective. If he sells pins in boxes of 100 and guarantees that not more than 10 pins will be defective, what is the approximate probability that a box will fail to meet the guaranteed quality?

Remarks

$$n = 100, p = .05 \Rightarrow np = 5$$

$$P(X > 10) = 1 - P(X \leq 10) = 1 - \sum_{k=0}^{10} \frac{e^{-5} 5^k}{k!} =$$

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

JAN. 2002

~~Ex~~  $P(X=5) = 0.0004$

Ex Suppose that a container contains 10,000 particles. The prob. that such a particle escapes from the container equals 0.0004. What is the prob. that more than 5 such escapes occur?

Sol:  $n = 10,000, p = 0.0004$

$$\lambda = np = 4. \quad \therefore P(X > 5) = 1 - P(X \leq 5)$$